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BACHELOR OF SCIENCE IN ELECTRICAL AND ELECTRONICS

ENGINEERING (BSEEE)

MICROPROCESSOR SYSTEMS

LAB REPORT:

8086 Microprocessor Trainer Experiment Report

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1. Introduction

The purpose of this experiment is to understand the working of the 8086 microprocessor by executing different programs on the 8086 Microprocessor Trainer Kit. The experiment involves 16-bit arithmetic operations, memory block transfer, and input-output operations using the 8255 Programmable Peripheral Interface (PPI). The experiments also include testing RAM and ROM functionality,.

2. Experiment Objectives

- Perform 16-bit addition, subtraction, and multiplication, and store the results in memory.
- Execute memory block transfer and fill operations.
- Configure and control input-output operations using the 8255 PPI.
- Test RAM and ROM functionality using checksum and RAM test routines.

3. Equipment Used

- 8086 Microprocessor Trainer Kit
- Power Supply
- Hexadecimal Keypad (for program entry)
- 7-Segment Display (for result observation)

4. Procedure & Observations

1. 16-Bit Addition Program

Program Entry:

- Enter the following program at memory location 00800h.

Address	Mnemonics	Opcode	Description
00800h	MOV AX, [00600h]	A1 00 26	Load first number into AX
00803h	MOV BX, [00602h]	8B 1E 02 26	Load second number into BX
00807h	ADD AX, BX	03 C3	Perform addition
00809h	MOV [00604h], AX	A3 04 26	Store result at 00604h
0080Ch	CALL ADDRDISP	9A A0	Call display subroutine

- Data Input:
 - Store first 16-bit number at 00600h (LSB) and 00601h (MSB).
 - Store second 16-bit number at 00602h (LSB) and 00603h (MSB).
- Execution & Result Verification:
 - Run the program from 00800h.
 - The result will be stored in 00604h (LSB) and 00605h (MSB).
 - Verify the result by checking the memory location and display.

2. 8255 Programmable Peripheral Interface (PPI) Experiment:

Ports and Addresses:

The 8255 PPI provides three 8-bit ports (Port A, Port B, and Port C), which can be configured as input or output based on control word programming. The port addresses are as follows:

Description	Port Address
Command port address of 8279	12h
Data port address of 8279	10h
Command port address of 8255	3Eh
PA Port address of 8255	38h
PB Port address of 8255	3Ah
PC Port address of 8255	3Ch

Programming Control Bytes

Control Byte	Port A	Port B	Port C0 - C3	Port C4 - C7
80h	OUT	OUT	OUT	OUT
81h	OUT	OUT	OUT	IN
82h	OUT	OUT	IN	OUT
83h	OUT	OUT	IN	IN
89h	OUT	IN	OUT	OUT
8Ah	OUT	IN	OUT	IN
8Bh	OUT	IN	IN	OUT
8Ch	OUT	IN	IN	IN
90h	IN	OUT	OUT	OUT
91h	IN	OUT	OUT	IN
92h	IN	OUT	IN	OUT

93h	IN	OUT	IN	IN
98h	IN	IN	OUT	OUT
99h	IN	IN	OUT	IN
9Ah	IN	IN	IN	OUT
9Bh	IN	IN	IN	IN

The 8086 Microprocessor Trainer is equipped with an 8255 PPI IC, which allows communication between the microprocessor and peripheral devices. The 8255 provides three 8-bit ports (Port A, Port B, and Port C), which can be configured as input or output based on control word programming.

- The trainer includes 24 programmable I/O lines.
- The trainer features battery backup to retain RAM contents in case of power failure.

3. Input-Output Control Using 8255 PPI

- **Configuring the Ports:**

- Write control word to configure Port A, B, and C for input/output.
- Ensure correct data flow based on control byte settings.

- **Testing Outputs:**

- Send test data to Port A and observe LED response.
- Verify expected behavior using an oscilloscope or display.

- **Testing Inputs:**

- Read input signals from switches and verify correct data reception.
- Ensure the microprocessor correctly interprets the input state.

4. Memory Block Transfer and Fill Operations

Memory Block Transfer:

- To transfer a block of memory, specify the source and destination addresses.
- Example: Transfer 5 bytes from 00900h to 00904h to 00500h.
- Use the MOVE/FILL key to enter the source and destination addresses.

Memory Fill Operation:

- To fill a block of memory with a specific value, use the FILL function.
- Example: Fill memory locations 00600h to 00610h with the value ABh.
- Use the ALT key to select the FILL function and enter the start and end addresses.

5. RAM Test / Checksum

- The RAM test function checks the integrity of the RAM by performing non-destructive read/write operations.
- The checksum function computes the checksum for ROM and displays the result.

5. Expected Results

The 16-bit addition program should correctly add two 16-bit numbers and store/display the result.

The 8255 PPI should correctly configure and control input-output ports based on the control byte settings.

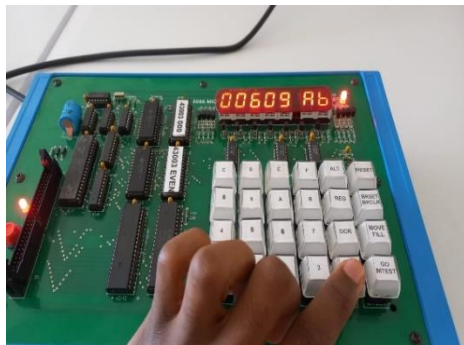
The memory block transfer and fill operations should successfully move and fill memory blocks as specified.

The RAM test and checksum functions should verify the integrity of RAM and ROM.

6. Results

16-bit addition: The addition operation executed successfully, and the sum was stored and accessed as expected.

Moving and filling block of memory: Proper execution of the control word program, allowing correct manipulation of ports.

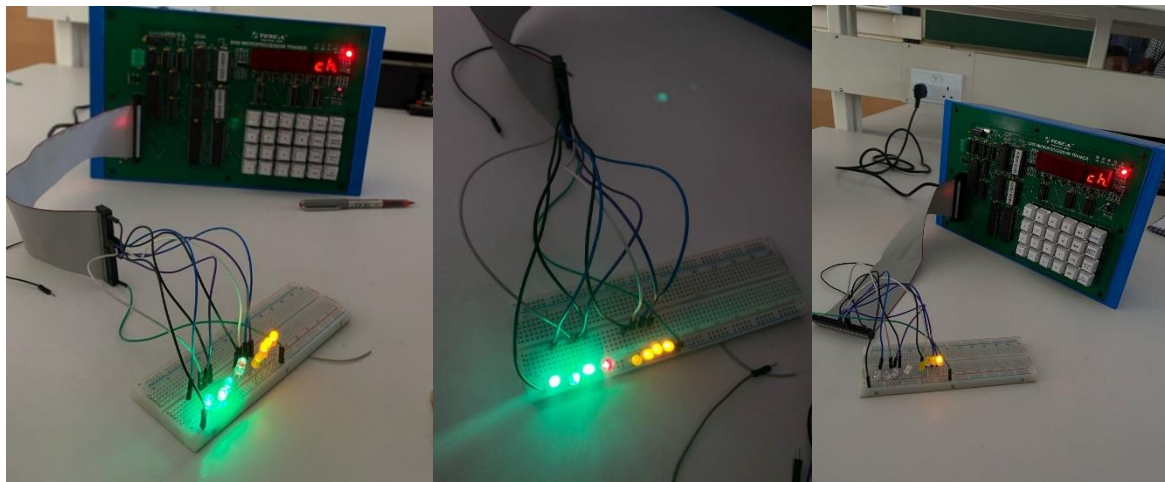




Programmable Peripheral Interface (PPI) Experiment:

Each LED corresponds to a single bit, where a lit LED represents a '1' and an unlit LED represents a '0'.

- When entering 01 (hex), only the least significant LED was lit.
- When entering FF (hex), all LEDs were lit, indicating an 8-bit representation of all ones (1111 1111 in binary).
- The experiment successfully demonstrated the correct binary output corresponding to each hexadecimal input.



RAM Test / Checksum

Test results matched expected checksum values.

7. Conclusion

This experiment successfully demonstrated various functionalities of the 8086 microprocessor, including arithmetic operations, memory manipulation, input-output control using the 8255 PPI, and waveform generation using the DAC module. Understanding these functionalities is essential for designing microprocessor-based systems in real-world applications.

Despite some difficulties with certain boards that did not work properly, the different experiments successfully demonstrated key operations of the 8086 microprocessor, including arithmetic computations, memory manipulations, and input-output control using the 8255 PPI. The execution of RAM integrity tests, memory block transfers, and modifications highlighted the importance of efficient memory management in microprocessor-based systems. Through these experiments, we gained valuable insight into assembly language programming and how microprocessor systems function.

8. Appendices

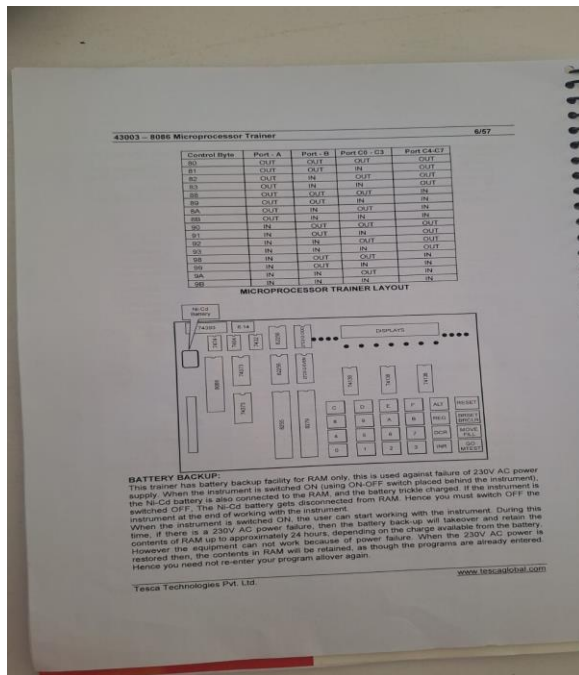


Figure 1. microprocessor trainer layout

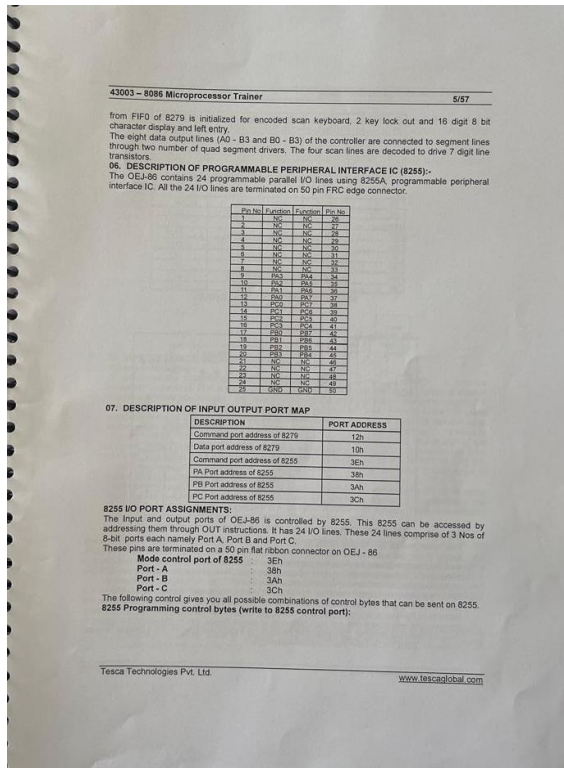


Figure 3 3. 16 bit addition

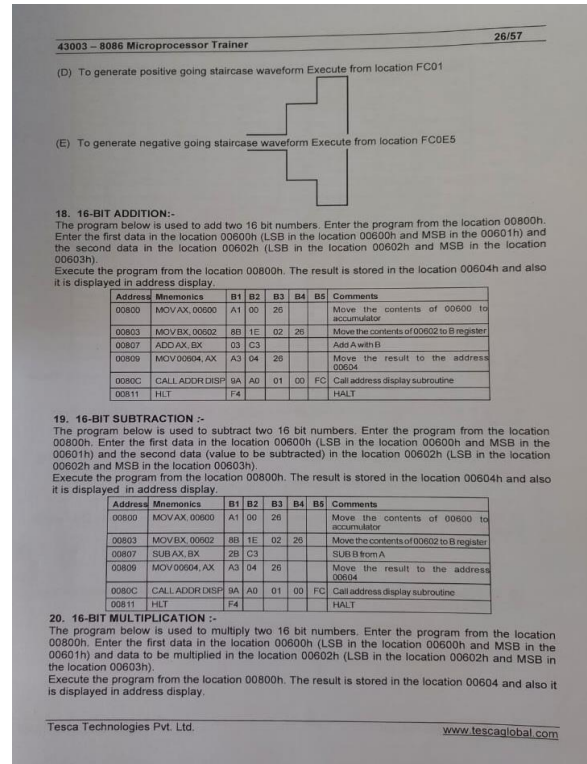


Figure 2 2. 16 bit addition

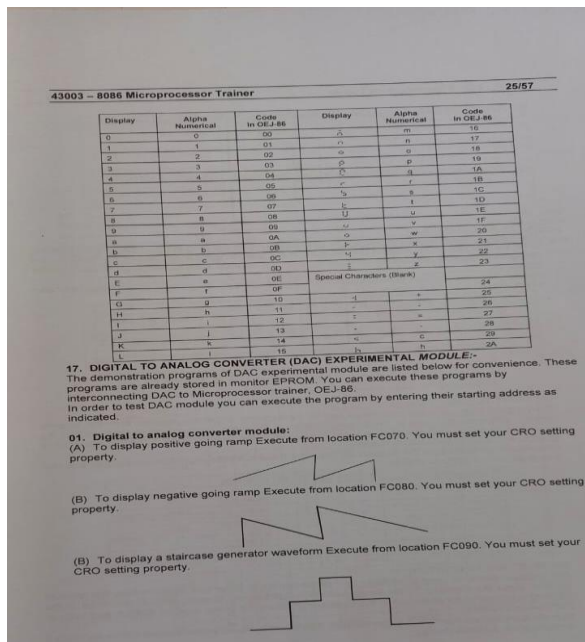


Figure 55. RAM test / checksum

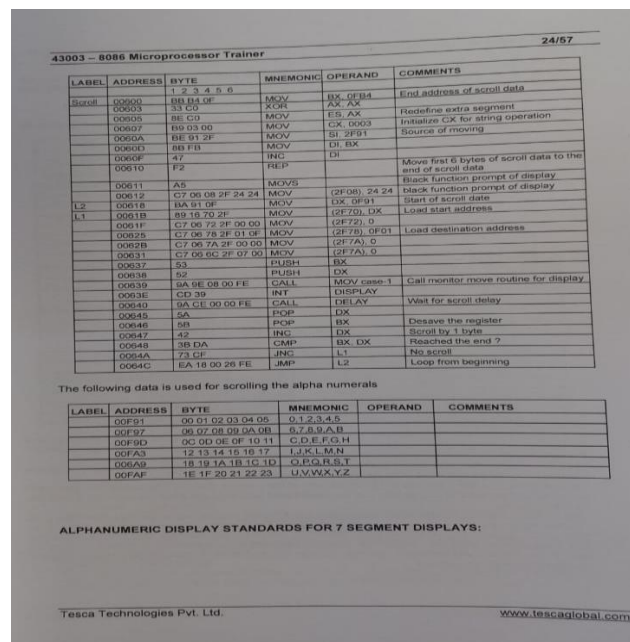


Figure 44. RAM test / checksum

